

Editors
Journal of Public Economics

Dear Professors Hendren and Kopczuk,

I am submitting “Quantifying the Annuity Puzzle: A Unified Lifecycle Decomposition” for consideration at the *Journal of Public Economics*.

Yaari (1965) proved that rational consumers should fully annuitize under actuarially fair pricing, yet voluntary ownership among US retirees is only a few percent. Six decades of research have proposed partial explanations, but the literature has lacked a unified quantitative accounting of how the main channels interact in a single model. Pashchenko (2013, *JPubE*) came closest, but still predicted participation of roughly 20%.

The paper develops two complementary models. **Model 1** is a structural lifecycle decomposition. A nine-channel rational-plus-preference specification (Social Security, bequests, medical expenditure risk with health-mortality correlation, survival pessimism, pricing loads, inflation erosion, long-term care risk, age-varying consumption needs from Aguiar–Hurst 2013, and state-dependent utility from Finkelstein–Luttmer 2013) predicts ownership of 6.8%. This sits modestly above both HRS empirical comparators reported in parallel—3.11% (95% CI [2.54%, 3.81%]) for the cleaner fat-file lifetime contract indicator and 5.21% (95% CI [4.45%, 6.09%]) for the conventional any-annuity income proxy used in prior literature, both computed on the same wealth-restricted sample as the model predictions. We do not claim that the model “matches” or “resolves” the puzzle; the structural channels alone bring predicted ownership into the same neighborhood as the HRS comparators, with the prediction running roughly 1.6 to 3.7 percentage points above the two empirical measures.

Model 1 extended. Layering two behavioral channels onto the nine-channel baseline yields an eleven-channel specification with predicted ownership of 0.0%. The behavioral channels are source-dependent utility (Blanchett–Finke 2024, 2025) and a purchase-event disutility consistent with the Brown (2008) / Chalmers–Reuter (2012) / Hu–Scott (2007) narrow-framing literature. Both are calibrated to literature magnitudes rather than estimated from US moments; we are explicit that the eleven-channel specification is exploratory and that the structural-baseline-plus-residual reading is the disciplined one.

Model 2 is a reduced-form transport using the UK 2015 pension-freedoms reform as out-of-sample variation. Pre-reform mandatory annuitization sustained DC-pot annuity retention of 95%; post-reform retention is 17% at the mid anchor (range 13%–25% across ABI and FCA industry estimates, supplemented by individual-level ELSA pension-grid evidence). Applied as a behavioral wedge against the frictionless population benchmark of 41.8%, this yields a predicted US ownership of 7.5% (bracket [5.7%, 11.0%]). The two models reach the same neighborhood from independent directions: structural decomposition from inside the lifecycle problem, and cross-country transport from outside it. The conventional HRS income proxy (5.21%, CI [4.45%, 6.09%]) sits just below the Model 2 bracket’s lower edge, with its upper confidence bound extending into the bracket; the cleaner lifetime-contract indicator falls below the bracket.

The paper makes four contributions. First, the unified structural framework—the first to nest

age-varying consumption needs alongside the established rational and preference channels of the annuity-puzzle literature. Second, the cross-validation of structural decomposition against an external reduced-form transport based on the UK 2015 reform, drawing on individual-level ELSA pension-grid records alongside industry aggregates. Third, an exact Shapley decomposition over all $2^9 = 512$ subsets of the nine structural channels (the disciplined headline attribution), with an exploratory eleven-channel extension over $2^{11} = 2,048$ subsets that adds the two literature-magnitude behavioral parameters; the behavioral channels enter with large but offsetting contributions (42.8 pp for purchase-event disutility, -30.3 pp for source-dependent utility) and are reported as illustrative rather than identified. Fourth, two distinct policy levers emerge: a supply-side reform (group pricing) raises predicted ownership relative to the structural baseline, and a demand-side reform anchored externally by Model 2 operates on a separate margin; neither is a clean structural identification of the underlying mechanism.

The paper is a good fit for the *Journal of Public Economics*. It engages the journal's core interests in Social Security, retirement saving, insurance-market frictions, and the design of public and employer-sponsored retirement institutions, and it engages directly with Pashchenko (2013) and related work published in *JPubE* and neighboring outlets.

Replication code in Julia and all calibration data are publicly available at <https://github.com/DerekTharp/annuity-puzzle-model>. The manuscript has not been submitted elsewhere.

Sincerely,

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